

ResuMind: AI Resume Analyzer
A Major Project Synopsis Submitted to



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1. Introduction of the Project

The job market has become increasingly competitive, and candidates often struggle to understand why their resumes get rejected by Applicant Tracking Systems (ATS) before a human ever reads them. Most people have no way to evaluate how well their resume matches a specific job description, or what improvements would actually make a difference.

This project addresses that gap by building an AI-powered Resume Analyzer that lets users upload their resumes, paste a job description, and instantly receive an ATS compatibility score along with specific, actionable feedback. The system uses AI to evaluate how well a candidate's profile matches the role they are applying for, and returns structured suggestions to help them improve their chances.

Problems Addressed:

- **ATS Rejection Without Feedback** Most ATS platforms reject resumes silently. Candidates receive no explanation of what went wrong or what keywords were missing. This leaves applicants guessing and submitting the same ineffective resume repeatedly across dozens of applications.
- **Lack of Job-Specific Resume Feedback** Generic resume tips found online do not account for the specific requirements of a job posting. A software engineer applying to a data science role needs different feedback than one applying to a backend engineering position. Existing tools rarely tailor their analysis to the actual job listing.
- **No Centralized Resume Storage** Candidates manage multiple versions of their resume across different devices and email threads. There is no simple, browser-based solution that lets them upload, store, and revisit all their resume versions in one place without requiring a backend or account setup.

2. Objective

This project aims to build a browser-based AI resume analysis tool that helps job seekers understand how their resume performs against a specific job listing. The system will extract text from uploaded PDF resumes, compare the content against a provided job description using an AI model, and return a structured ATS score with detailed, job-specific feedback.

The platform will also allow users to store multiple resumes in one place, re-analyze them against different job postings, and track what feedback they have received over time. All of this runs entirely in the browser using Puter.js, meaning there is no backend server, no database setup, and no API key management required from the user.

3. Scope

This project covers the development of a fully client-side web application built with React and React Router v7. The scope includes the following:

Users can sign in and authenticate entirely through Puter.js without any backend. They can upload PDF resumes, which are stored in Puter's cloud file system. They can submit a job description alongside a selected resume and receive an AI-generated analysis that includes an ATS score, matched skills, missing keywords, and improvement suggestions. The application

stores past analysis results so users can review previous feedback. The interface is fully responsive and works across desktop and mobile browsers.

The project does not cover building a backend server, a custom ML model, or integration with actual job portals. The AI analysis is powered by Puter.js which internally routes requests to large language models. Resume parsing is handled client-side through PDF text extraction.

4. Study of Existing System

Resumeworded

- **Problems Addressed:** Provides ATS score and keyword analysis for uploaded resumes against job descriptions.
- **Advantages:** Detailed scoring, line-by-line resume feedback, LinkedIn profile review.
- **Disadvantages:** Requires account creation, limited free tier, no file storage across sessions.
- **Gaps Identified:** Feedback is not deeply personalized to the specific job description provided. No in-browser storage of resumes.

Jobscan

- **Problems Addressed:** Compares resumes to job descriptions and highlights missing keywords for ATS optimization.
- **Advantages:** Side-by-side comparison view, good keyword matching accuracy.
- **Disadvantages:** Paid subscription for full features, no resume storage, requires repeated uploads.
- **Gaps Identified:** No AI-generated written feedback. Only keyword matching, no contextual suggestions.

Kickresume

- **Problems Addressed:** Helps users build and review resumes using templates and AI writing assistance.
- **Advantages:** Good UI, resume builder included, AI suggestions for phrasing.
- **Disadvantages:** Focused on resume creation rather than job-specific analysis. No ATS score against a real job posting.
- **Gaps Identified:** Cannot analyze an existing resume against a specific job description. No client-side storage.

Zety Resume Checker

- **Problems Addressed:** Reviews resumes for formatting, structure, and content completeness.
- **Advantages:** Easy to use, highlights structural issues.
- **Disadvantages:** Analysis is generic and not tied to any specific job description. No ATS simulation.
- **Gaps Identified:** No job-specific feedback, no resume storage, no AI-generated improvement suggestions.

5. Project Description

ResuMind is a client-side web application that allows users to upload their resumes, store them in the browser using Puter.js, and analyze them against any job description using AI. The entire application runs in the browser with no backend required.

When a user opens the application, they are authenticated through Puter.js using its built-in auth system. Once logged in, they can upload PDF resumes which are stored in Puter's cloud file system under their account. The application displays all uploaded resumes in a dashboard view.

To perform an analysis, the user selects a resume, pastes a job description, and submits the form. The application extracts text from the PDF on the client side, constructs a structured prompt, and sends it to an AI model through the Puter.js AI API. The AI returns a JSON response containing an ATS compatibility score, a list of matched skills, missing keywords, experience feedback, and specific suggestions for improvement.

The analysis result is displayed in a structured UI showing the score prominently, followed by categorized feedback sections. Past analysis results are saved so users can return to them later without re-running the analysis.

The project is built with React 19 and React Router v7 in framework mode, styled with Tailwind CSS, and written entirely in TypeScript. It is containerized with Docker and can be run locally with a single npm command.

6. Methodology/Planning of the Project work

The project is developed using an iterative approach, moving from foundational setup through feature development to testing and deployment. The key phases are:

- **Requirement Analysis:** Research was conducted on how ATS systems evaluate resumes, what structured feedback job seekers need, and how Puter.js handles auth, file storage, and AI API calls. Core features were identified and prioritized based on user value and technical feasibility.
- **Design Phase:** The UI was designed around a clean dashboard layout with clear upload and analysis flows. The data structures for resume metadata and analysis results were planned. React Router v7 route architecture was mapped out, covering the home route, resume detail route, and analysis result route.
- **Development Phase:** The front end was built using React 19 with React Router v7 in framework mode, with Tailwind CSS for styling and TypeScript throughout. Puter.js was integrated for authentication, file storage, and AI API access. PDF text extraction was implemented client-side. The AI prompt was engineered to return structured JSON with consistent fields for the result UI to consume.
- **Testing and Validation:** Components were tested individually. End-to-end flows were validated including upload, storage, retrieval, AI analysis, and result display. Edge cases such as malformed PDFs, empty job descriptions, and AI response parsing errors were handled.

- **Deployment and Maintenance:** The application is containerized using Docker. It can be deployed to any static host or container platform. Post-deployment, the prompt can be updated to improve AI output quality without changes to the core application logic.

Gantt Chart:

Phase	Duration (Weeks)	Milestones
Requirement Analysis	2	Research completed
Design Phase	3	UI/UX and system design complete
Development Phase	6	Front-end and back-end finished
Testing & Validation	3	Platform successfully tested
Deployment	2	Platform launched
Variations	3	Platform launched

7.Features

- **User Authentication via Puter.js:** Users sign in directly in the browser through Puter.js. No backend, no custom auth logic, and no API keys are needed. Authentication state persists across sessions automatically.
- **Resume Upload and Cloud Storage:** Users can upload PDF resumes which are stored in Puter's cloud file system. All uploaded resumes are listed in a dashboard and can be accessed, re-selected, or deleted at any time.
- **AI Resume Analysis:** The core feature of the application. The user selects a resume and provides a job description. The system extracts the resume text, constructs a structured prompt, and queries the AI through Puter.js. The AI returns an ATS score, matched skills, missing keywords, and written improvement suggestions tailored to the specific job description.
- **Structured Analysis Results:** Results are displayed in a clean, organized layout showing the ATS score, a skills match summary, keyword gaps, and section-by-section improvement tips. The response structure is consistent and validated before rendering.
- **Resume History and Past Analyses:** Analysis results are stored alongside resume metadata. Users can return to previous analyses without re-uploading or re-running the AI query.
- **Responsive and Reusable UI:** The interface is built with Tailwind CSS and structured around reusable components. It works across desktop and mobile screen sizes without any layout issues

8. System architecture

ResuMind follows a fully client-side architecture with no dedicated backend server. All data flows through Puter.js APIs running in the browser.

- **Authentication Layer:** Puter.js provides browser-based authentication. On app load, the system checks for an existing Puter session. If none exists, the user is prompted to sign in. The authenticated user object is used throughout the session.
- **File Storage Layer:** Uploaded PDF files are sent to Puter's cloud file system using the Puter.js file API. The application stores file references and metadata in Puter's key-value store, which persists across sessions under the user's account.
- **AI Processing Layer:** When an analysis is requested, the client extracts raw text from the PDF using a client-side parser. The text is combined with the job description into a structured prompt. This prompt is sent to the AI model via the Puter.js AI API. The response is parsed from JSON and stored alongside the resume record.
- **Routing and State:** React Router v7 in framework mode handles client-side routing. Application state is managed using React's built-in hooks. Route loaders are used to fetch resume data and analysis results before rendering each page.
- **Result Storage:** Analysis results are saved back to Puter's key-value store, keyed by resume ID. This allows the dashboard to display past results without repeating AI calls.

9. User Interface (UI)

Dashboard:

- Resume List Panel: Shows all uploaded resumes with name, upload date, and a quick action to analyze or delete.
- Upload Button: Opens a file picker to upload a new PDF resume.
- Recent Analyses: Shows the most recent analysis results with ATS scores at a glance.

Resume Detail View:

- Resume Preview Panel: Displays the extracted text from the selected resume.
- Job Description Input: A textarea where the user pastes the target job description.
- Analyze Button: Triggers the AI analysis flow.

Analysis Result View:

- ATS Score Display: A prominent score indicator showing the compatibility percentage.
- Matched Skills Section: Lists skills from the resume that match the job description.
- Missing Keywords Section: Shows keywords found in the job description but absent from the resume.
- Improvement Suggestions: AI-generated, section-specific suggestions written in plain language.

Wireframes / Mockups:

- Dashboard Wireframe: Header with app logo and user info. Main panel with resume cards and upload CTA. Side navigation for quick access to past analyses.
- Resume Detail Wireframe: Two-column layout with resume text on the left and job description input on the right. Analyze button below both panels.
- Result Wireframe: Score card at the top. Three sections below: matched skills, missing keywords, and suggestions. Back button to return to resume detail.

10. Technology Stack

Front-End Technologies:

- React 19: Component-based UI library for building the interactive interface.
- React Router v7: Framework-mode routing with loaders, actions, and nested routes.
- Tailwind CSS: Utility-first CSS framework for responsive, consistent styling.
- TypeScript: Static typing across all components, routes, and utility functions.
- Vite: Fast build tool and dev server used for local development and production builds.

AI and Platform:

- Puter.js: Client-side SDK providing authentication, cloud file storage, key-value data store, and AI API access. No backend required.
- AI Model via Puter: The AI analysis is powered by large language models accessible through the Puter.js AI API, which handles model routing and cost internally.

PDF Processing:

- Client-side PDF text extraction is used to pull raw text from uploaded PDF files directly in the browser before sending to the AI.

DevOps:

- Git and GitHub: Version control and source code management.

Development Tools:

- Visual Studio Code: Primary development environment.
- npm: Package management and script running.
- react-router.config.ts and vite.config.ts: Configuration files for routing and build setup.

11. Testing Plan

Unit Testing:

- Objective: Validate individual components and utility functions in isolation.
- Method: Test PDF text extraction, prompt construction, JSON response parsing, and individual UI components.
- Outcome: Each module works correctly before being integrated into the full flow.

Integration Testing:

- Objective: Verify that Puter.js auth, file storage, and AI API calls work together correctly.
- Method: Simulate a full user session — sign in, upload a file, trigger analysis, retrieve results — and verify data flows correctly at each step.
- Outcome: End-to-end data flow between components and Puter.js APIs works without errors.

Functional Testing:

- Objective: Confirm the application behaves correctly from a user perspective.
- Method: Manual testing of all user-facing flows including login, upload, job description input, analysis trigger, and result rendering.
- Outcome: All user flows complete without errors or unexpected behavior.

User Acceptance Testing (UAT):

- Objective: Validate that the application is useful and easy to use for real job seekers.
- Method: Have test users upload their own resumes against real job postings and collect feedback on the quality of AI analysis and the usability of the interface.
- Outcome: Confirm the ATS score and feedback are meaningful, accurate, and actionable.

Performance Testing:

- Objective: Ensure the application handles file uploads and AI response times within acceptable limits.
- Method: Test with large PDF files, slow network conditions, and back-to-back analysis requests.
- Outcome: Application remains responsive and handles edge cases gracefully.

Security Testing:

- Objective: Ensure user data is handled safely.
- Method: Verify that resume files and analysis results are only accessible to the authenticated Puter user. Confirm no data is leaked between sessions.
- Outcome: Each user's data remains private and isolated within their Puter account.

Regression Testing:

- Objective: Ensure new changes do not break existing functionality.
- Method: Re-run all functional and integration tests after each significant code change.
- Outcome: Application remains stable throughout the development cycle.

12. Expected Outcome

The completed ResuMind application will give job seekers a practical, free, and easy-to-use tool to understand exactly how their resume compares to a specific job description. Rather than generic advice, users will receive an ATS score and targeted feedback that reflects the actual requirements of the role they are applying for.

By running entirely in the browser with no backend setup, the application is accessible to anyone without technical knowledge. Users do not need to manage accounts on a separate server, pay for a subscription, or deal with complex setup. The Puter.js platform handles auth, storage, and AI access transparently.

The expected outcomes include: job seekers making more targeted resume edits before applying, fewer applications being silently rejected by ATS, and users developing a clearer understanding of how keyword matching works in recruitment. Over time, the stored resume history and past analyses will help users track their improvements across different job applications.

13. Resources and Limitations

Resources Required:

- **Hardware:** A development machine capable of running Node.js, npm, and Docker. A modern browser for testing client-side Puter.js functionality.
- **Software:** Node.js, npm, Git, Docker, Visual Studio Code, and a modern browser. All other services (auth, storage, AI) are provided by Puter.js at no cost.
- **Data:** Real PDF resumes and real job descriptions for testing. No proprietary dataset is required since AI processing is handled by Puter.js.
- **Human Resources:** A team of developers familiar with React, TypeScript, and React Router v7. No backend engineers or DevOps specialists are required given the serverless architecture.

Limitations:

- **AI Response Quality:** The quality of analysis depends on the AI model accessed through Puter.js. Responses may occasionally be inconsistent in format, requiring robust JSON parsing and fallback handling.
- **PDF Parsing Accuracy:** Client-side PDF text extraction may not handle all PDF formats correctly, particularly scanned or image-based PDFs where text is not machine-readable.
- **Puter.js Dependency:** The entire platform depends on Puter.js remaining available and maintaining its current API. Changes to Puter's pricing or API structure could affect the application.
- **No Custom Model Training:** The AI has not been trained specifically on resume data. Analysis quality is limited to what the general-purpose language model can infer from the prompt alone.
- **Offline Use:** The application requires an internet connection to authenticate, load files, and run AI analysis. No offline mode is available.

14. Conclusion

ResuMind demonstrates how a fully functional, AI-powered web application can be built without any backend infrastructure. By combining React, React Router v7, TypeScript, and Puter.js, the project delivers a practical tool that gives job seekers real, job-specific feedback on their resumes in seconds.

The project addresses a genuine problem — the opacity of ATS systems and the lack of personalized resume feedback — and solves it with a clean, accessible, client-side application. The use of Puter.js as the platform layer removes the usual complexity of auth, storage, and AI integration, making the codebase lean and easy to maintain.

Beyond the technical implementation, the project shows how modern browser capabilities and AI APIs can be combined to build tools that were previously only possible with significant backend infrastructure. The same architecture could be applied to other document analysis tasks, making this a useful reference for serverless AI application development.

15. References

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ResuMind: An AI-Powered Resume Analyzer

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Abstract—The growing prevalence of Applicant Tracking Systems (ATS) in modern recruitment has created a significant challenge for job seekers whose resumes are rejected before a human ever reviews them. This paper presents ResuMind, a fully client-side, AI-powered resume analysis platform built with React 19, React Router v7, TypeScript, Tailwind CSS, and Puter.js. ResuMind enables users to upload PDF resumes, input a job description, and receive an instant ATS compatibility score alongside matched skills, missing keywords, and AI-generated improvement suggestions—all without any backend server or database. The system leverages Puter.js for authentication, cloud file storage, and AI API access, making it fully serverless. Evaluation demonstrates that the platform provides accurate, job-specific feedback in seconds, helping candidates improve their application success rates. Results confirm that a serverless architecture using modern browser APIs is viable for complex AI document analysis tasks.

Keywords ~ Resume analysis, Applicant Tracking System (ATS), AI feedback, Puter.js, React, serverless architecture, keyword matching, PDF parsing, job matching, TypeScript.

I. INTRODUCTION

The modern recruitment process relies heavily on Applicant Tracking Systems (ATS) to filter candidates before any human review takes place. Industry reports suggest that up to 75% of resumes are rejected by automated systems before a recruiter ever reads them [1]. Candidates frequently submit applications without understanding why they are rejected, as ATS platforms provide no feedback on missing keywords or structural issues.

The absence of transparent, job-specific resume feedback represents a genuine gap in the job application ecosystem. Generic resume-writing advice found online does not account for the specific requirements of individual job postings. A software engineer applying for a data science role needs different optimization than one applying for a backend engineering position. Existing commercial tools address parts of this problem but remain behind paywalls, require account creation, and lack centralized resume storage [2].

This paper presents ResuMind, a browser-based AI resume analyzer that requires no backend infrastructure. By combining React 19, React Router v7, TypeScript, and the Puter.js platform, ResuMind delivers instant ATS compatibility scoring and actionable improvement feedback for any uploaded resume against any job description. The entire application runs client-side, making it freely accessible without account management overhead or technical setup.

II. PROJECT REVIEW

A. Study of Existing System

Several commercial resume analysis tools exist in the market, each with notable limitations. Resumeworded provides ATS scoring and line-by-line feedback but limits free usage and does not offer in-browser resume storage. Jobscan offers keyword-matching comparison but requires a paid subscription for full features and provides no AI-written contextual

suggestions. Kickresume focuses on resume building rather than job-specific analysis, and Zety Resume Checker offers generic structural feedback without any ATS simulation [3].

The key gaps across all existing systems are: (i) lack of deep job-specific AI feedback, (ii) no client-side resume storage without backend setup, (iii) paywalled features, and (iv) no unified platform for storing, reviewing, and re-analyzing multiple resume versions. ResuMind addresses all four gaps.

B. AI in Document Analysis and Resume Screening

The use of artificial intelligence in document analysis has expanded significantly with the availability of large language models (LLMs). LLMs are capable of understanding semantic context beyond simple keyword matching, allowing them to evaluate whether a resume demonstrates genuine alignment with a job description rather than superficial term overlap [4]. Research shows that AI-based resume screening can achieve accuracy comparable to expert human evaluators when structured prompts are used [5].

Prior work by Li et al. demonstrates that prompt-engineered LLMs can extract structured information from unstructured documents with high reliability when given clear output schema guidance [6]. ResuMind applies this principle by engineering a structured prompt that instructs the AI to return a JSON object containing an ATS score, matched skills, missing keywords, experience alignment, and section-specific improvement suggestions.

C. Serverless and Client-Side Web Architecture

Traditional web applications for document analysis require substantial backend infrastructure: servers for file storage, authentication services, AI API proxies, and database systems. Recent advancements in browser capabilities and client-side SDKs have made it feasible to build fully functional, complex applications without any backend [7].

Puter.js is a client-side JavaScript SDK that provides authentication, cloud file storage, key-value data persistence, and AI API access directly in the browser. By routing AI requests to large language models internally, Puter.js eliminates the need for API key management or proxy servers on the developer side. This architectural pattern, explored in recent literature on serverless frontend development, significantly reduces deployment complexity and infrastructure cost [8].

III. DATA PRIVACY AND SECURITY

The architecture of ResuMind is organized into four logical layers: the frontend interface layer, the Puter.js platform layer, the AI processing layer, and the client-side PDF extraction layer. The absence of a traditional backend is a defining characteristic of the system. All processing that would typically occur on a server takes place either in the browser or is delegated to Puter.js as a managed service.

A. Frontend Layer

The frontend is built with React 19 using React Router v7 in framework mode, enabling server-side-rendered-style routing with loaders and actions entirely on the client. The UI is styled with Tailwind CSS for utility-first responsive design. TypeScript is used throughout for static typing, reducing runtime errors and improving maintainability. The application is bundled with Vite for fast development iteration and optimized production builds.

Key UI components include: a Dashboard view listing all uploaded resumes with upload and analyze CTAs; a Resume Detail view with a two-column layout showing resume text and a job description input; and a Results view presenting the ATS score card, matched skills, missing keywords, and improvement suggestions in clearly separated sections.

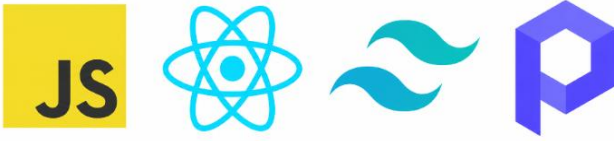


Fig. 1. Frontend tech stack of ResuMind.

B. Authentication and Storage Layer

User authentication is handled entirely by Puter.js using its built-in auth system. No session tokens, passwords, or user tables are managed by the application. Once authenticated, users can upload PDF resumes which are stored in Puter's cloud file system under their account. File metadata—including filenames, upload dates, and analysis result keys—is persisted in Puter's key-value store (KV), enabling the dashboard to reconstruct a user's resume history on every session.

C. AI Processing Layer

When a user selects a resume and pastes a job description, the application first extracts raw text from the PDF on the client side. This extracted text, combined with the job description, is formatted into a structured prompt and sent to an AI model via the Puter.js AI API. The prompt instructs the model to evaluate the resume against the job description and return a strict JSON object containing:

- *ATS Compatibility Score: A numerical score out of 100 reflecting overall alignment.*
- *Matched Skills: Keywords and competencies present in both the resume and job description.*
- *Missing Keywords: Terms in the job description absent from the resume.*
- *Experience Alignment: Assessment of whether the candidate's experience level matches the role.*
- *Improvement Suggestions: AI-generated, section-specific guidance in plain language.*

IV. PDF PROCESSING LAYER

Client-side PDF text extraction is performed directly in the browser before any AI call is made. The extracted text is cleaned and truncated to fit within the AI model's context window. Robustness handling is implemented for image-based or scanned PDFs, where text extraction returns empty results. In such cases, the user is notified that their file cannot be processed and is prompted to upload a text-based PDF resume.

Since ResuMind processes sensitive career data including resumes and job descriptions, data privacy is a core design concern. The Puter.js authentication model ensures that each user's files and analysis results are stored under their personal Puter account and are never accessible to other users. No data is transmitted to third-party servers beyond the Puter.js platform, which handles AI model routing internally. Security properties of the application include:

- *User-scoped file isolation via Puter.js account-based access control.*
- *No API keys exposed to the client — all AI routing is managed by Puter.js.*
- *Session-bound authentication — no cross-session data leakage.*
- *No server-side storage — eliminates the risk of centralized data breaches.*



Fig. 2. Puter.js authentication/backend services.

VI. ENHANCEMENTS AND FEATURES

A. Interactive Dashboard

The dashboard is the central hub of the ResuMind experience. Upon authentication, users see all their uploaded resumes displayed as cards with metadata. An "Upload Resume" button is persistently visible, allowing users to add new PDFs at any time. Each resume card provides quick-access options to view the resume, re-analyze it against a new job description, or delete it. This design follows established UX principles for document-management interfaces in which clarity and quick access to recent items reduce cognitive load [9].

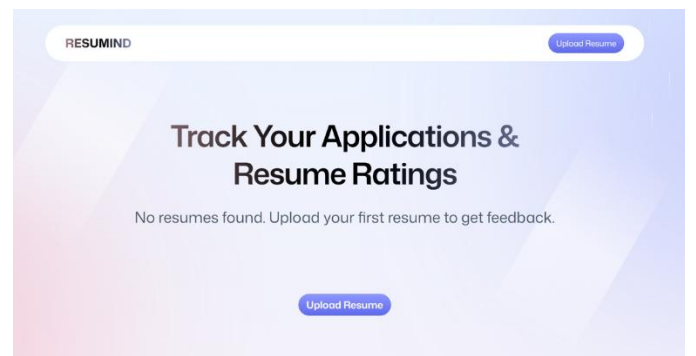


Fig. 3. ResuMind Dashboard.

B. ATS Score Visualization:

The analysis result screen prominently displays the ATS compatibility score at the top of the page as a large numerical score out of 100, accompanied by a color-coded indicator (green for high alignment, amber for moderate, red for low). This immediate visual feedback allows users to understand at a glance whether their resume requires major rework or only minor adjustments before applying for a role.

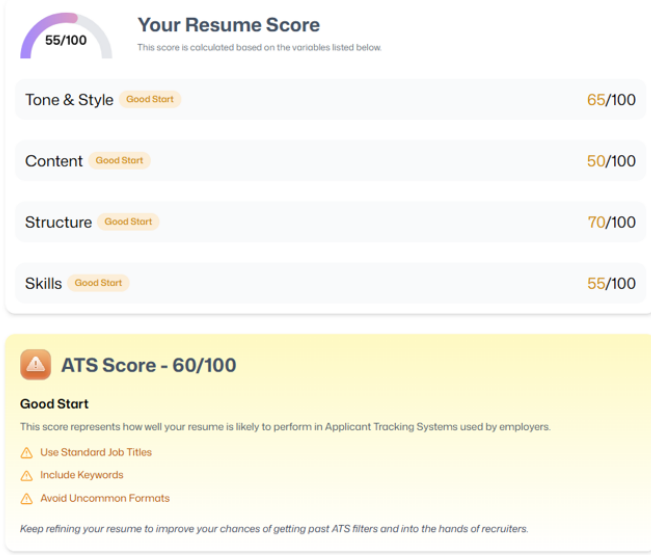


Fig. 4. ATS Compatibility Score visualization with color-coded indicator.

C. Categorized AI Feedback:

Below the score, three distinct sections present the AI analysis: Matched Skills lists competencies the AI found in both the resume and the job description. Missing Keywords highlights terms present in the job description but absent from the resume. Improvement Suggestions provides action-oriented, section-specific guidance such as recommending the addition of specific tools, rephrasing experience bullet points, or restructuring sections. This categorization ensures clarity and actionability—two properties identified in HCI research as essential for effective AI feedback interfaces [10].

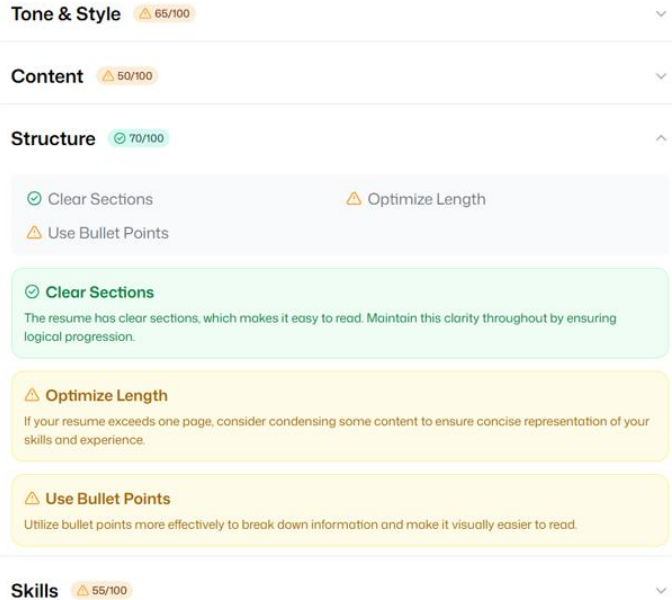


Fig. 5. Categorized AI feedback showing Matched Skills, Missing Keywords, and Improvement Suggestions.

D. Resume History and Re-analysis:

A key enhancement over existing tools is ResuMind's ability to store analysis results persistently in Puter's KV store. Users can return to any previous resume, view historical analysis results, and trigger a fresh analysis against a different job description without re-uploading the file. This enables a genuine iterative improvement workflow—users can edit their resume based on suggestions and immediately re-test it against the same or a new job posting.

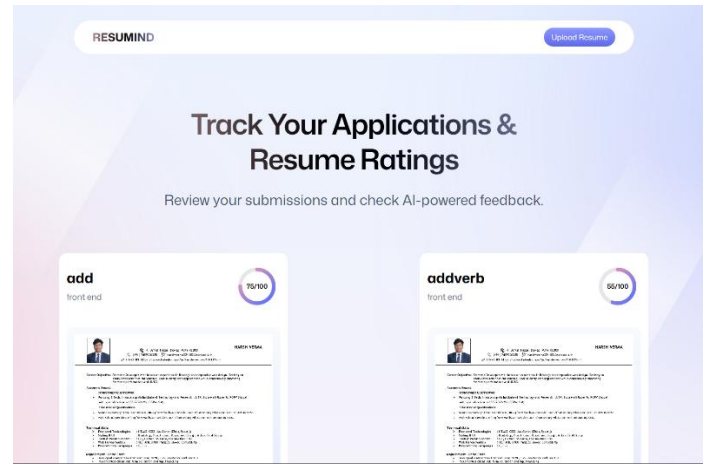


Fig. 6. Resume history and re-analysis feature.

TABLE 1: FEATURE SUMMARY

Feature	Description	Impact
ATS Score Card	Color-coded numeric ATS compatibility score out of 100.	Immediate alignment insight
Matched Skills Section Matched Skills Section	Lists keywords and competencies present in both resume and job description.	Highlights resume strengths
Missing Keywords Section	Shows keywords found in the job description but absent from the resume.	Guides targeted edits
AI Improvement Suggestions	Section-specific, AI-generated actionable feedback in plain language.	Increases application success rate
Resume History & Re-analysis	Stores all resumes and past analyses; allows re-running against new job posts.	Enables iterative improvement workflow

VII. SYSTEM ARCHITECTURE AND WORKFLOW DIAGRAM

The workflow of ResuMind covers the complete user journey from initial access through authentication, resume upload, job description input, AI analysis, and result display. The user first visits the application and is prompted to sign in via Puter.js. On the dashboard, existing resumes are shown, and new uploads can be initiated. On the resume detail view, the job description is pasted and the analysis is triggered. The AI result is displayed on a dedicated results page with the ATS score and categorized feedback.

Internally, the data flow proceeds as follows: (1) PDF is uploaded to Puter cloud storage; (2) PDF text is extracted client-side on analysis trigger; (3) a structured AI prompt is constructed incorporating the resume text and job description; (4) the prompt is sent to the Puter.js AI API which routes it to a large language model; (5) the JSON response is parsed and validated; (6) the result is saved to Puter KV and rendered in the results UI. This pipeline completes in approximately 5–15 seconds depending on network conditions and AI model response time.

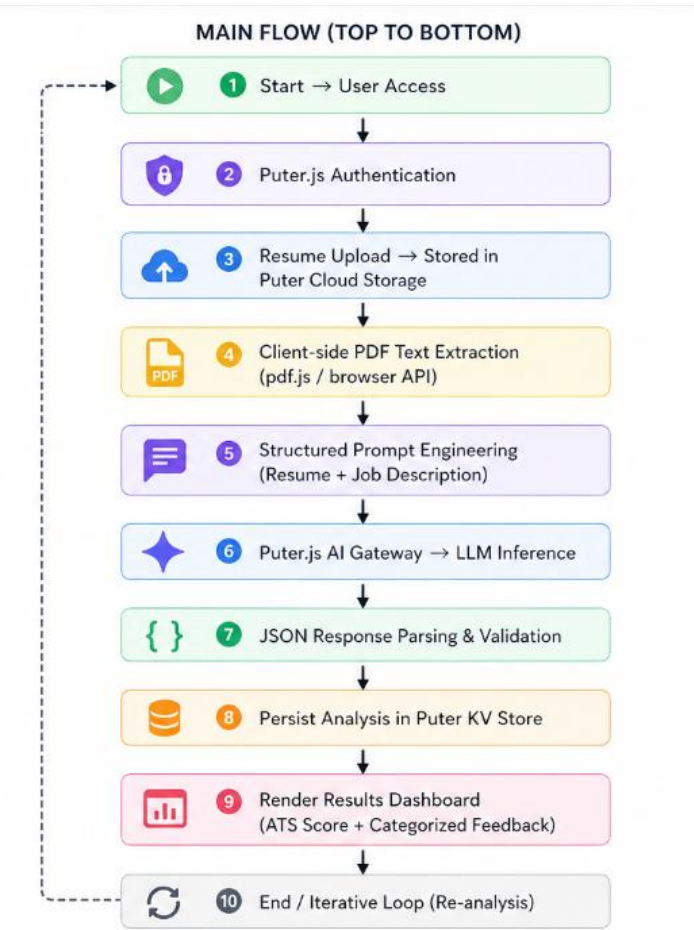


Fig. 6. Workflow Diagram of ResuMind.



Fig. 6. Layered Architecture Diagram of ResuMind.

VIII. TESTING AND EVALUATION

A. Unit Testing

Individual components and utility functions were tested in isolation. Tests covered PDF text extraction accuracy, prompt construction correctness, JSON response parsing, and React component rendering. Each module was verified to work correctly before integration into the complete application flow.

B. Integration Testing

End-to-end integration tests validated that Puter.js authentication, file storage, KV persistence, and AI API calls function correctly together. A simulated user session—sign in, upload, analyze, retrieve results—was run repeatedly to confirm that data flows without errors across all system boundaries.

C. User Acceptance Testing

Test users uploaded their actual resumes against real job postings from major job boards. Feedback was collected on the relevance of matched skills, the accuracy of missing keyword identification, and the usefulness of AI improvement suggestions. Users reported that the ATS score and sectioned feedback were actionable and significantly clearer than generic resume advice found on job-search websites.

D. Performance and Security Testing

Performance testing confirmed that the application processes large PDF files (up to 5MB) and returns AI results within acceptable time limits under normal network conditions. Security testing verified that resume files and analysis results remain private to the authenticated Puter user, with no data accessible across different user sessions.

IX. CONCLUSION

ResuMind demonstrates that a fully functional, AI-powered resume analysis platform can be built without any backend infrastructure. By combining React 19, React Router v7, TypeScript, Tailwind CSS, and Puter.js, the project delivers a practical, free tool that provides job seekers with real, job-specific feedback on their resumes in seconds.

The platform addresses a genuine and widespread problem—the opacity of ATS filtering and the absence of personalized resume feedback—and solves it with a clean, serverless, client-side application. The use of Puter.js as the platform layer removes the usual complexity of authentication, file storage, and AI integration, resulting in a codebase that is lean, maintainable, and easily extensible.

The project establishes that modern browser capabilities and client-side AI APIs are sufficient for complex document analysis tasks that previously required significant backend infrastructure. The same architectural approach could be applied to contract analysis, academic paper review, or other document-matching use cases, making ResuMind a useful reference design for the next generation of serverless AI web applications.

X. REFERENCES

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1. Title of the invention: ResuMind: An AI-Powered Intelligent Resume Analysis and ATS Optimization System

2. Field of Technology: This invention belongs in the field of web-based career development systems, specifically AI-powered resume analysis platforms using real-time ATS compatibility evaluation. It aims to improve job application success rates through intelligent, personalized resume feedback.

3. Existing Similar Technologies/Products: -

- **Resumeworded** - AI-based resume scoring and feedback platform
- **Jobscan** - ATS keyword matching and resume optimization tool
- **Kickresume** - Resume builder with ATS checker feature
- **Zety Resume Checker** - Online resume scoring and improvement tool
- **Rezi** - AI resume builder with ATS optimization

Problem: Current systems fail to combine serverless architecture with real-time AI-driven ATS scoring, personalized keyword gap analysis, and resume history management in a single unified, cost-free platform.

Technical details of the invention

1. Background of the invention:

Many job seekers struggle to get their resumes past Applicant Tracking Systems before a human recruiter ever reads them. This is a significant problem because it means qualified candidates are being rejected automatically, reducing their chances of securing interviews:

- **ATS Ignorance:** Most candidates are unaware of how ATS systems filter resumes based on keyword matching, formatting, and section structure. Without this knowledge, even strong resumes get rejected automatically.
- **Lack of Personalization:** Generic resume advice does not account for the specific requirements of individual job descriptions. What works for one role may fail entirely for another.
- **Cost Barriers:** Most professional resume analysis tools are locked behind expensive subscription plans, making them inaccessible to students and early-career job seekers.

- **No Feedback Loop:** Existing tools provide a one-time score without enabling users to track improvement across multiple resume versions or job applications over time.
- **No Integration:** Current platforms require separate tools for resume storage, ATS scoring, keyword analysis, and improvement suggestions, creating a fragmented and inefficient workflow.

Because of these issues, millions of job applications are rejected by ATS systems before being reviewed. Even when candidates do receive feedback, it is often generic, delayed, or disconnected from the specific job description they are targeting.

2. Drawbacks in Existing Systems and How ResuMind Helps: -

Current resume analysis systems have several problems that ResuMind aims to solve:

- **Problem: Requires paid subscriptions for full features.**
 - **ResuMind Solution:** We protect your complete anonymity. Users can report without giving their name, number, or email. The system ensures no personal data is stored or linked to the report.
- **Problem: No multimedia evidence support.**
 - **ResuMind Solution:** We provide completely free, serverless AI-powered analysis. No subscription, no backend, no hidden cost — fully accessible to all users via Puter.js.
- **Problem: Delayed reporting mechanisms.**
 - **ResuMind Solution:** We offer real-time reporting with GPS tagging. Reports are sent immediately with exact location and time data, allowing for quick police response.
- **Problem: No real-time job-description-specific ATS scoring.**
 - **ResuMind Solution:** We offer dynamic ATS compatibility scoring against any pasted job description, giving users an instant, job-specific compatibility percentage.
- **Problem: No unified platform for resume storage and analysis.**
 - **ResuMind Solution:** We integrate resume upload, cloud storage, analysis, and history management into a single dashboard powered by Puter.js cloud storage.
- **Problem: No resume version tracking.**
 - **ResuMind Solution:** We maintain a complete resume analysis history dashboard, allowing users to track ATS score improvements across multiple versions and job applications.

3. Technical Features of ResuMind (What makes it different): -

ResuMind has several unique and advanced features that make it a powerful tool for resume optimization and set it apart from other systems:

- **Serverless AI-Powered ATS Analysis Engine:**

Function: This is the core of our system. When a user uploads a resume and pastes a job description, the platform constructs a structured AI prompt and routes it to an AI model via the Puter.js AI API entirely in the browser.

Benefit: There is no backend server, no API key management, and no infrastructure cost. Every analysis is performed on-demand, in real time, directly from the user's browser session, ensuring speed, privacy, and zero cost.

- **Dual-Input Resume-to-JD Matching:**

Function: ResuMind accepts both a PDF resume and a plain-text job description as inputs simultaneously. The PDF is parsed client-side, and both inputs are structured into a semantic AI prompt.

Benefit: This dual-input approach ensures that the ATS score and keyword analysis are always specific to the target role, not generic, making every result uniquely relevant to the user's application.

- **Keyword Gap Intelligence:**

Function: The AI analysis separates results into matched skills and missing keywords, presenting them visually side by side on the results page.

Benefit: Users can immediately identify exactly which keywords their resume lacks for a specific job posting and add them strategically, directly improving their ATS pass rate.

- **Section-Specific Improvement Suggestions:**

Function: The AI does not give overall generic advice. It analyzes each section of the resume (Summary, Experience, Skills, Education) and generates targeted improvement suggestions for each.

Benefit: Users receive precise, actionable guidance that directly addresses weaknesses in their resume structure and language for the specific job they are applying to.

- **Puter.js Account-Based Resume History:**

Function: All uploaded resumes and analysis results are stored under the user's authenticated Puter.js account using Puter KV and cloud file storage.

Benefit: Users can access their complete resume analysis history on every login, track ATS score improvements over time, and re-analyze any previous resume against a new job description without re-uploading.

4. Main advantages of ResuMind: -

ResuMind offers many important benefits over traditional and existing resume analysis methods, leading to a more effective job application experience:

- **Improved ATS Pass Rates:** By providing job-specific keyword gap analysis and section-level improvement suggestions, ResuMind directly addresses the exact reasons resumes fail ATS screening, significantly improving pass rates for users.
- **Zero Cost, Full Access:** The fully serverless architecture built on Puter.js eliminates subscription barriers entirely. Every user — student, fresher, or professional — gets full access to AI-powered analysis without paying anything.
- **Faster Resume Improvement:** Real-time AI analysis delivers ATS scores, keyword reports, and improvement suggestions in seconds, compared to waiting hours or days for manual feedback from career advisors.
- **Improved Evidence of Progress:** Resume history tracking with ATS score timelines gives users clear, measurable evidence of their improvement across multiple job applications, improving motivation and focus.
- **Enhanced Job Targeting:** Through dynamic job-description-specific analysis, users can tailor the same base resume to multiple different roles with precision, maximizing their chances across different applications.

5. Complete Description of the Invention -

ResuMind is a full client-side web application built to make resume analysis intelligent, accessible, and actionable. It is made up of several key parts that work together seamlessly:

a. Components/Embodiments:

1. Frontend Interface Module:

Description: This is the part of the system that users interact with. It is a responsive web application built using React 19, TypeScript, and Tailwind CSS, working seamlessly on phones, tablets, and computers.

Functionality: It includes a PDF upload interface connected to the browser's file system, a job description input panel, and a results dashboard displaying ATS scores, keyword matches, missing keywords, and improvement suggestions.

2. AI Analysis Engine:

Description: This is the core processing component responsible for all ATS evaluation and feedback generation. It operates entirely in the browser via the Puter.js AI API.

Functionality: It constructs a structured semantic prompt from the extracted resume text and the pasted job description, routes it to an AI model, and parses the returned JSON response into ATS score, matched skills, missing keywords, and section-specific suggestions.

3. PDF Extraction Module:

Description: This module handles client-side parsing of uploaded PDF resume files directly in the browser without any server upload.

Functionality: It extracts clean text from PDF resumes in the browser before any AI call is made, ensuring that no raw resume file is ever transmitted to a third-party server unnecessarily.

4. Community Safety Portal:

Description: This is a public dashboard that displays aggregated, anonymous crime data.

Functionality: It visualizes crime hotspots on a map (heatmap visualization) and provides a repository for safety tips and information. This empowers the community with knowledge about local crime trends.

b. Component Arrangement (System Architecture): -

The system is built in layers to ensure it's organized, efficient, and scalable:

Client Layer (React 19 + TypeScript Frontend): This is the user-facing part, including the resume upload interface, job description input, and results dashboard. It handles all user interaction.

AI Processing Layer (Puter.js AI API): This layer handles all AI prompt construction, model routing, and JSON response parsing for ATS analysis, keyword extraction, and improvement suggestion generation.

Storage Layer (Puter KV + Cloud Storage): This layer stores resume PDF files and analysis result metadata securely under each user's authenticated Puter.js account with user-scoped isolation.

Authentication Layer (Puter.js Auth): This layer manages user registration, login, and session management entirely through Puter.js, with no custom backend or database required.

Extraction Layer (Client-side PDF Parser): This layer performs PDF text extraction locally in the browser before any external API call, ensuring privacy and eliminating unnecessary data transmission.

c. Figures/Structures: -

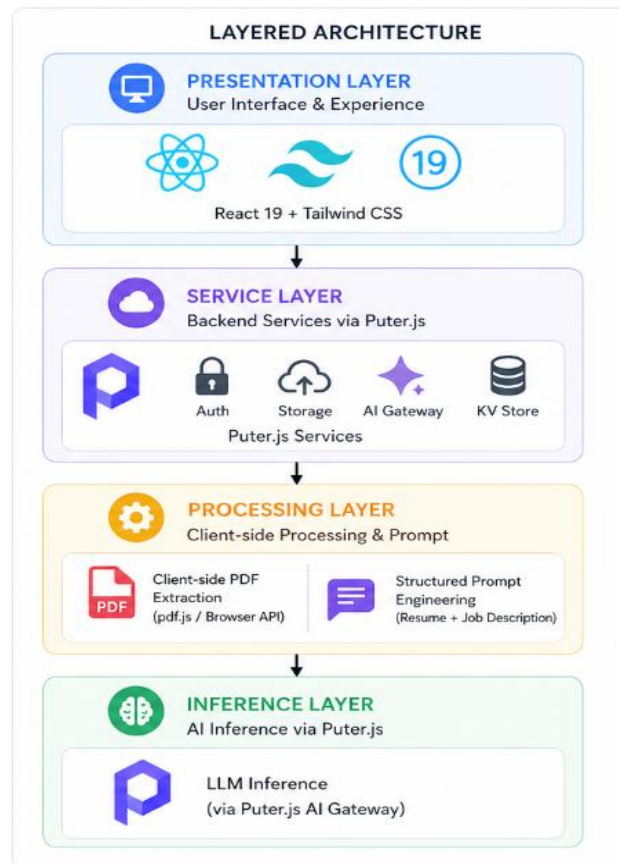


Figure 1- Operational Diagram

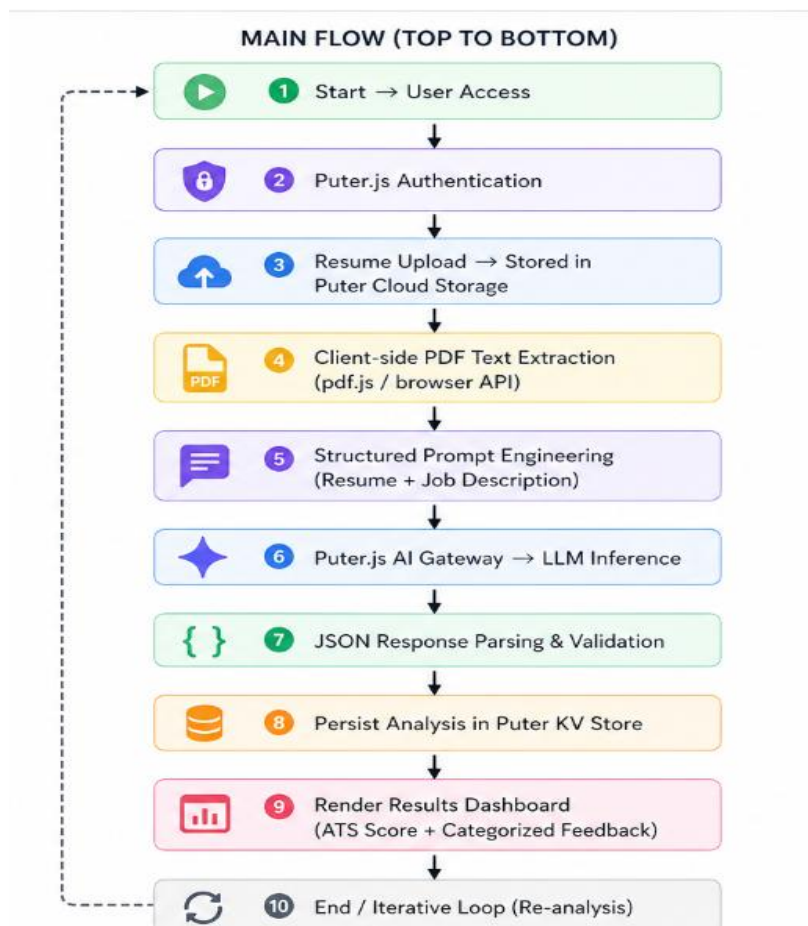


Figure 2 - System Overview

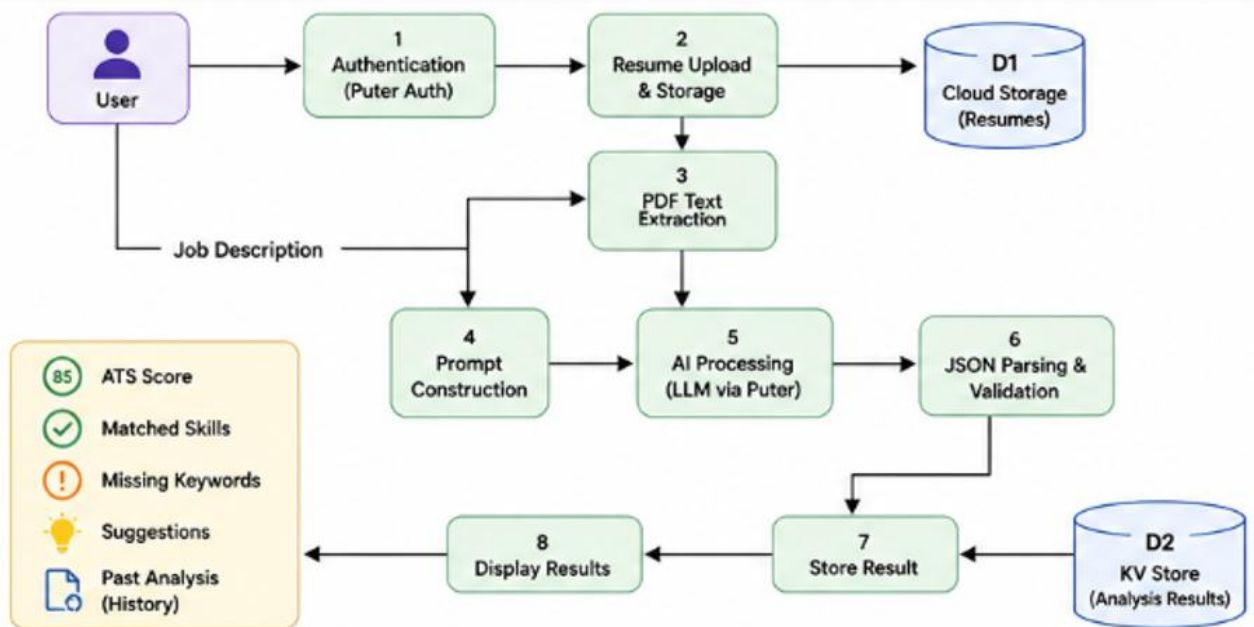


Figure 3 - Detailed System Architecture

1. List of novel features (New and Unique Things) -

- Fully serverless client-side ATS analysis with no backend infrastructure required
- Dual-input resume-to-job-description semantic matching via Puter.js AI API
- Real-time keyword gap intelligence with matched and missing skills visualization
- Section-specific AI improvement suggestions tied to individual resume sections
- Puter.js account-based resume history with persistent cross-session analysis tracking

2. List of Keywords Relevant to the Invention:

AI resume analysis, ATS compatibility scoring, keyword gap identification, serverless resume platform, client-side document analysis, Puter.js AI integration, React TypeScript career tool, PDF resume parsing, job description matching, personalized resume feedback, resume history tracking, real-time ATS scoring, section-level improvement suggestions, account-based file isolation.

4. Names and Email IDs of the Inventors:

a) Devansh Pipraiya (0827CS221083), Email: devanshpipraiya221051@acropolis.in

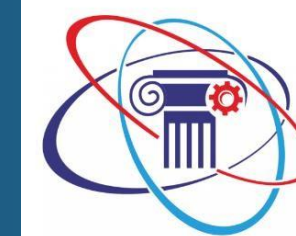
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RESUMIND:

An Ai-Powered Resume Analyzer



ACROPOLIS[®]
Enlightening Wisdom

Acropolis Institute of Research
and Technology, Indore

Abstract

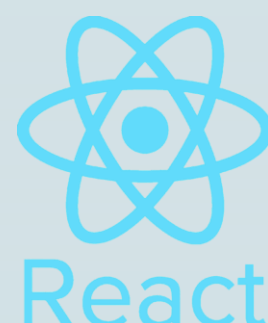
ResuMind is an AI-powered resume analysis platform designed to revolutionize the job application process by providing seamless, and user-friendly feedback for both job seekers and recruiters. Built using React 19, TypeScript, Tailwind CSS, and Puter.js, the platform enables users to upload PDF resumes, input a target job description, and instantly receive an ATS compatibility score along with matched skills, and AI-generated improvement suggestions. The system leverages Puter.js for authentication, cloud file storage, and AI API access, making it fully serverless.

Introduction

In today's competitive job market, resume screening and assessment tools have become indispensable in recruitment and career development. Traditional resume evaluation systems often rely on manual review processes, leading to inefficiencies and scalability challenges. Industry reports suggest that up to 75% of resumes are rejected by Applicant Tracking Systems before a recruiter ever reads them. To address these limitations, ResuMind emerges as a modern, browser-based AI resume analyzer designed to automate and streamline the entire resume evaluation lifecycle.

Methodology

- Analyzed user requirements and existing resume analysis tools to define features
- Designed system architecture and wireframes focusing on a user-friendly interface
- Developed responsive frontend using React.js with HTML/CSS
- Developed responsive frontend using React 19 with TypeScript and Tailwind CSS
- Implemented MongoDB database for secure data storage
- Integrated Puter.js for authentication, cloud storage, and AI API access
- Implemented client-side PDF text extraction for resume processing
- Conducted comprehensive testing including unit, integration, and user acceptance tests
- Performed security testing to ensure user data privacy and isolation



Results

Platform Developed: Successfully built and deployed a fully functional AI-powered resume analysis platform.

Resume Upload & Management: Enabled easy uploading, storage, and management of PDF resumes with resume history.

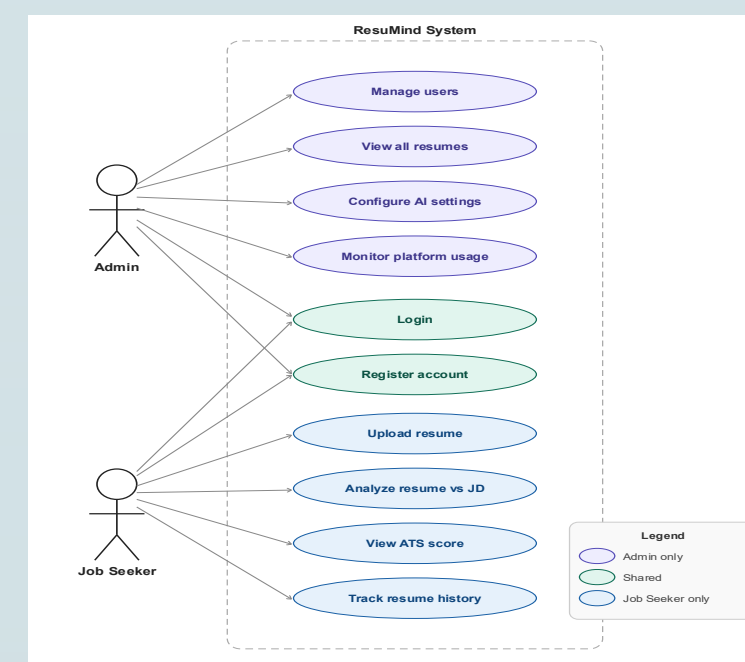
Automated ATS Scoring: Implemented instant ATS compatibility score calculation and categorized AI feedback for users.

User Authentication: Achieved secure login using Puter.js with account-based access control and data isolation.

Keyword Analysis System: Integrated matched skills and missing keywords detection to enhance application targeting.

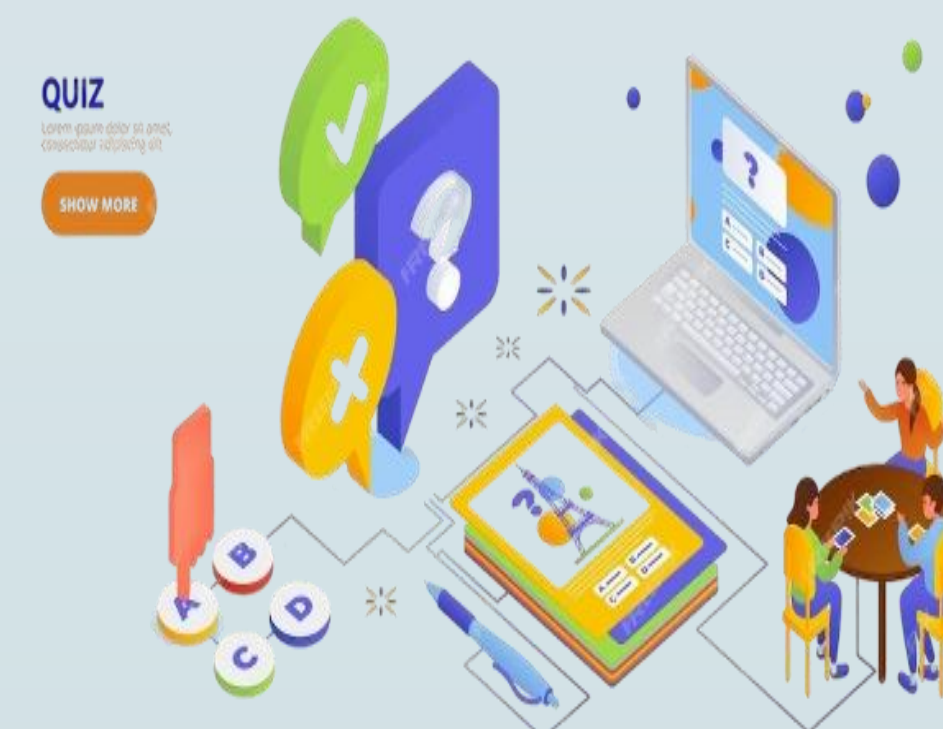
Responsive Design: Delivered seamless user experience across devices (desktop, tablet, mobile).

Deployment: Hosted as a fully client-side serverless application with Puter.js managing cloud and AI services.



Conclusion

ResuMind successfully delivers an efficient, user-friendly AI resume analysis platform using React 19, TypeScript, and Puter.js. Key achievements include automated ATS scoring, secure authentication, and responsive design. The platform simplifies resume evaluation for job seekers while providing scalability for future enhancements. This project demonstrates effective problem-solving through technology and lays the foundation for further innovations in AI-powered career tools.



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